

1) List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case

Computer won't power on – A multimeter checks the power supply's output voltages to confirm whether the PSU is delivering the correct power or has failed.

Smartphone won't charge – A multimeter measures the charger, USB cable, or battery voltage to identify whether the charging accessories or battery are faulty.

Laptop or smartphone has a suspected broken connection – A multimeter's continuity mode tests cables, connectors, or fuses to detect open circuits or damaged connections.

**2) Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.
Hint: Focus on the codes shown and how they relate to hardware issues.**

Watch This Video: - <https://www.youtube.com/watch?v=ApLWyORilVc>

A POST card displays **hexadecimal POST codes** while the motherboard starts, showing which stage of hardware initialization has been reached.

In the demonstration, the card's code changes during boot and stops at a particular code when the motherboard encounters a problem.

The code is then compared with the motherboard's POST-code table to identify the faulty area, such as **RAM, CPU, or another hardware component**.

This makes the POST card useful for locating hardware faults when a computer fails to boot or gives no display.

3) Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.

Constraint: Blur or crop out any personal information before submitting.

HWMonitor				
File View Tools Help				
Sensor	Value	Min	Max	
TCPL-6-1				
Dell Inc. 02N3WF				
Temperatures				
TZ.TZ00	28.0 °C	28.0 °C	28.0 °C	
Fans				
Fan 0	0 RPM	0 RPM	0 RPM	
Utilization				
Physical Memory Load	91.0 %	90.0 %	91.0 %	
Available Physical Mem...	0.7 GB	0.7 GB	0.7 GB	
Virtual Memory Load	63.6 %	63.5 %	63.6 %	
Available Virtual Memory	4.8 GB	4.8 GB	4.8 GB	
Intel Core i3 8100T				
Voltages				
VID (Max)	0.978 V	0.896 V	0.978 V	
IA Offset	+0.000 V	+0.000 V	+0.000 V	
GT Offset	+0.000 V	+0.000 V	+0.000 V	
Ring Offset	+0.000 V	+0.000 V	+0.000 V	
System Agent Offset	+0.000 V	+0.000 V	+0.000 V	
Temperatures				
Cores (Max)	59.0 °C	59.0 °C	65.0 °C	
Package	62.0 °C	59.0 °C	64.0 °C	
Powers				
Package	8.64 W	1.37 W	8.64 W	
IA Cores	8.36 W	1.15 W	8.36 W	
GT	0.02 W	0.01 W	0.14 W	
DRAM	0.58 W	0.33 W	0.58 W	
Power Max (PL1)	35.00 W	35.00 W	35.00 W	
Short Power Max (PL2)	60.00 W	60.00 W	60.00 W	
Max Peak Power (PL4)	63.00 W	63.00 W	63.00 W	
Currents				
IccMax	104.00 A	104.00 A	104.00 A	
Clocks				
Cores (Max)	3092.4 MHz	3090.9 MHz	3092.9 MHz	
CPU BCLK	99.8 MHz	99.7 MHz	99.8 MHz	
LLC/Ring	2793.2 MHz	2791.8 MHz	2793.2 MHz	
Memory	1197.1 MHz	1197.1 MHz	1197.1 MHz	
Utilization				
Processor	7.5 %	4.4 %	27.2 %	
ST1000LM035-1RK172 (WKPTE8V3)				
Temperatures				
Assembly	49.0 °C	49.0 °C	49.0 °C	
Air Flow	49.0 °C	49.0 °C	49.0 °C	
Counters				
Power-On-Hours	10954 hrs	10954 hrs	10954 hrs	
Power Cycle Count	3315	3315	3315	
Utilization				
Used Space (e:)	88.1 %	88.1 %	88.1 %	
Available Capacity (e:)	11.6 GB	11.6 GB	11.6 GB	

Ready

4) Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it checks	Example Problem Detected
Multimeter	Measures voltage, resistance, and electrical continuity.	Faulty power supply or broken charging cable.
POST Card	Displays motherboard POST/diagnostic codes during startup.	Faulty RAM, CPU, or motherboard preventing the computer from booting.
Software Diagnostics	Checks hardware and system performance using diagnostic programs.	Hard-drive errors, memory problems, or driver/software issues.